

DGX Spark (GB10) thermal-defect telemetry — raw logs

Host spark-5208 · DGX OS 7.2.3 · kernel 6.17.0-1031-nvidia · driver 580.173.02 · captured 2026-08-26. Thermal zone values are millidegrees C (e.g. 91200 = 91.2°C). GPU triple is power draw, die temperature, utilization.

results/thermal_verify.log

Controlled reproduction: sustained matmul load with an automatic safety abort at 88°C. Platform zones ran 79.6→89.4°C in 44 seconds at ≤97W while the GPU die read 64–77°C.

```
=== thermal_verify start 2026-08-26T17:15:15
 11s maxzone=79.6C gpu=[97.34 W, 67, 96 %]
 22s maxzone=83.8C gpu=[94.36 W, 64, 82 %]
 33s maxzone=86.8C gpu=[94.93 W, 75, 82 %]
 44s maxzone=89.4C gpu=[92.19 W, 77, 96 %]
SAFETY ABORT at 89.4C
VERDICT: RUNAWAY peak=89.4C elapsed=44s
```

results/thermal_governor.log

Interim mitigation log: SIGSTOP above 85°C / SIGCONT below 78°C. Note the pause/resume cadence — seconds-scale heating and cooling, consistent with a localized heat-transfer defect.

```
2026-08-26T16:22:34+01:00 PAUSE at 85500mC (pid 33664)
2026-08-26T16:22:44+01:00 RESUME at 69100mC (pid 33664)
2026-08-26T16:23:14+01:00 PAUSE at 86900mC (pid 33664)
2026-08-26T16:23:24+01:00 RESUME at 67700mC (pid 33664)
2026-08-26T16:23:44+01:00 PAUSE at 86000mC (pid 33664)
2026-08-26T16:23:54+01:00 RESUME at 67700mC (pid 33664)
2026-08-26T16:24:04+01:00 PAUSE at 85200mC (pid 33664)
2026-08-26T16:24:14+01:00 RESUME at 66400mC (pid 33664)
2026-08-26T16:24:34+01:00 PAUSE at 85700mC (pid 33664)
2026-08-26T16:24:44+01:00 RESUME at 68400mC (pid 33664)
2026-08-26T16:25:04+01:00 PAUSE at 86500mC (pid 33664)
2026-08-26T16:25:15+01:00 RESUME at 71800mC (pid 33664)
2026-08-26T16:25:25+01:00 PAUSE at 86000mC (pid 33664)
2026-08-26T16:25:35+01:00 RESUME at 70700mC (pid 33664)
2026-08-26T16:25:55+01:00 PAUSE at 88100mC (pid 33664)
2026-08-26T16:26:05+01:00 RESUME at 69400mC (pid 33664)
2026-08-26T16:26:15+01:00 PAUSE at 85200mC (pid 33664)
2026-08-26T16:26:25+01:00 RESUME at 68400mC (pid 33664)
2026-08-26T16:26:45+01:00 PAUSE at 87700mC (pid 33664)
2026-08-26T16:26:55+01:00 RESUME at 70100mC (pid 33664)
2026-08-26T16:27:05+01:00 PAUSE at 86900mC (pid 33664)
2026-08-26T16:27:15+01:00 RESUME at 69700mC (pid 33664)
2026-08-26T16:27:35+01:00 PAUSE at 85700mC (pid 33664)
2026-08-26T16:27:45+01:00 RESUME at 70200mC (pid 33664)
2026-08-26T16:27:55+01:00 PAUSE at 87100mC (pid 33664)
2026-08-26T16:28:05+01:00 RESUME at 69900mC (pid 33664)
2026-08-26T16:28:25+01:00 PAUSE at 88400mC (pid 33664)
2026-08-26T16:28:35+01:00 RESUME at 71200mC (pid 33664)
2026-08-26T16:28:45+01:00 PAUSE at 85500mC (pid 33664)
2026-08-26T16:29:05+01:00 RESUME at 77700mC (pid 33664)
2026-08-26T16:29:15+01:00 PAUSE at 86900mC (pid 33664)
2026-08-26T16:29:35+01:00 RESUME at 77000mC (pid 33664)
2026-08-26T16:29:55+01:00 PAUSE at 89000mC (pid 33664)
2026-08-26T16:30:05+01:00 RESUME at 77500mC (pid 33664)
2026-08-26T16:30:15+01:00 PAUSE at 85700mC (pid 33664)
2026-08-26T16:30:25+01:00 RESUME at 70800mC (pid 33664)
2026-08-26T16:30:36+01:00 PAUSE at 86100mC (pid 33664)
2026-08-26T16:30:56+01:00 RESUME at 76500mC (pid 33664)
2026-08-26T16:31:16+01:00 PAUSE at 88400mC (pid 33664)
2026-08-26T16:31:36+01:00 RESUME at 70300mC (pid 33664)
2026-08-26T16:31:56+01:00 PAUSE at 87800mC (pid 33664)
2026-08-26T16:32:06+01:00 RESUME at 69700mC (pid 33664)
2026-08-26T16:32:26+01:00 PAUSE at 86800mC (pid 33664)
2026-08-26T16:32:36+01:00 RESUME at 74800mC (pid 33664)
2026-08-26T16:32:46+01:00 PAUSE at 92600mC (pid 33664)
2026-08-26T16:33:06+01:00 RESUME at 76600mC (pid 33664)
2026-08-26T16:33:16+01:00 PAUSE at 86300mC (pid 33664)
2026-08-26T16:33:36+01:00 RESUME at 77000mC (pid 33664)
2026-08-26T16:33:56+01:00 PAUSE at 87500mC (pid 33664)
2026-08-26T16:34:06+01:00 RESUME at 68500mC (pid 33664)
2026-08-26T16:34:26+01:00 PAUSE at 87000mC (pid 33664)
2026-08-26T16:34:36+01:00 RESUME at 70400mC (pid 33664)
2026-08-26T16:34:46+01:00 PAUSE at 86400mC (pid 33664)
2026-08-26T16:34:56+01:00 RESUME at 69400mC (pid 33664)
2026-08-26T16:35:16+01:00 PAUSE at 88700mC (pid 33664)
2026-08-26T16:35:26+01:00 RESUME at 70600mC (pid 33664)
2026-08-26T16:35:36+01:00 PAUSE at 87600mC (pid 33664)
2026-08-26T16:35:46+01:00 RESUME at 70000mC (pid 33664)
2026-08-26T16:35:57+01:00 PAUSE at 85600mC (pid 33664)
2026-08-26T16:36:07+01:00 RESUME at 69100mC (pid 33664)
2026-08-26T16:36:17+01:00 PAUSE at 86400mC (pid 33664)
2026-08-26T16:36:27+01:00 RESUME at 68800mC (pid 33664)
2026-08-26T16:36:37+01:00 PAUSE at 87600mC (pid 33664)
2026-08-26T16:36:47+01:00 RESUME at 68300mC (pid 33664)
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2026-08-26T16:36:57+01:00 PAUSE at 87200mC (pid 33664)
2026-08-26T16:37:07+01:00 RESUME at 68700mC (pid 33664)
2026-08-26T16:37:17+01:00 PAUSE at 85700mC (pid 33664)
2026-08-26T16:37:27+01:00 RESUME at 68500mC (pid 33664)
2026-08-26T16:37:37+01:00 PAUSE at 87000mC (pid 33664)
2026-08-26T16:37:47+01:00 RESUME at 69400mC (pid 33664)
2026-08-26T16:37:57+01:00 PAUSE at 86800mC (pid 33664)
2026-08-26T16:38:07+01:00 RESUME at 77000mC (pid 33664)
2026-08-26T16:38:17+01:00 PAUSE at 86600mC (pid 33664)
2026-08-26T16:38:27+01:00 RESUME at 70200mC (pid 33664)
2026-08-26T16:38:37+01:00 PAUSE at 86400mC (pid 33664)
2026-08-26T16:38:47+01:00 RESUME at 68800mC (pid 33664)
2026-08-26T16:38:57+01:00 PAUSE at 85600mC (pid 33664)
2026-08-26T16:39:07+01:00 RESUME at 74100mC (pid 33664)
2026-08-26T16:39:17+01:00 PAUSE at 85800mC (pid 33664)
2026-08-26T16:39:27+01:00 RESUME at 72100mC (pid 33664)
2026-08-26T16:39:37+01:00 PAUSE at 87700mC (pid 33664)
2026-08-26T16:39:47+01:00 RESUME at 72600mC (pid 33664)
2026-08-26T16:39:57+01:00 PAUSE at 87500mC (pid 33664)
2026-08-26T16:40:08+01:00 RESUME at 76600mC (pid 33664)
2026-08-26T16:40:18+01:00 PAUSE at 87400mC (pid 33664)
2026-08-26T16:40:28+01:00 RESUME at 72000mC (pid 33664)
2026-08-26T16:40:38+01:00 PAUSE at 87200mC (pid 33664)
2026-08-26T16:40:48+01:00 RESUME at 70500mC (pid 33664)
2026-08-26T16:40:58+01:00 PAUSE at 86400mC (pid 33664)
2026-08-26T16:41:08+01:00 RESUME at 70200mC (pid 33664)
2026-08-26T16:41:18+01:00 PAUSE at 86900mC (pid 33664)
2026-08-26T16:41:28+01:00 RESUME at 68800mC (pid 33664)
2026-08-26T16:41:38+01:00 PAUSE at 85900mC (pid 33664)
2026-08-26T16:41:48+01:00 RESUME at 68900mC (pid 33664)
2026-08-26T16:41:58+01:00 PAUSE at 85900mC (pid 33664)
2026-08-26T16:42:08+01:00 RESUME at 68300mC (pid 33664)
2026-08-26T16:42:18+01:00 PAUSE at 86700mC (pid 33664)
2026-08-26T16:42:28+01:00 RESUME at 68100mC (pid 33664)
2026-08-26T16:42:38+01:00 PAUSE at 85100mC (pid 33664)
2026-08-26T16:42:48+01:00 RESUME at 67400mC (pid 33664)
2026-08-26T16:43:08+01:00 PAUSE at 86800mC (pid 33664)
2026-08-26T16:43:18+01:00 RESUME at 68700mC (pid 33664)
2026-08-26T16:43:38+01:00 PAUSE at 86300mC (pid 33664)
2026-08-26T16:43:48+01:00 RESUME at 69400mC (pid 33664)
2026-08-26T16:44:08+01:00 PAUSE at 88200mC (pid 33664)
2026-08-26T16:44:18+01:00 RESUME at 69700mC (pid 33664)
2026-08-26T16:44:39+01:00 PAUSE at 88500mC (pid 33664)
2026-08-26T16:44:49+01:00 RESUME at 69900mC (pid 33664)
2026-08-26T16:44:59+01:00 PAUSE at 85100mC (pid 33664)
2026-08-26T16:45:09+01:00 RESUME at 68900mC (pid 33664)
2026-08-26T16:45:29+01:00 PAUSE at 87800mC (pid 33664)
2026-08-26T16:45:39+01:00 RESUME at 69800mC (pid 33664)
2026-08-26T16:45:59+01:00 PAUSE at 88400mC (pid 33664)
2026-08-26T16:46:09+01:00 RESUME at 70100mC (pid 33664)
2026-08-26T16:46:29+01:00 PAUSE at 88400mC (pid 33664)

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results/blackbox.log

Black-box recorder: 30-second fsync'd samples of thermal zones, GPU power/temp/util, load, memory. Recording began 2026-08-26 ~15:20; the 15:48:24 sample (zones peaking 91.2°C) is the final record before the instant power-off — the system journal's last line is a routine cron entry at 15:45:01.

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=== 2026-08-26T15:17:50+01:00
45500 43800 43800 43800 44400 45500 44700 (thermal mC)
11.38 W, 43, 0 %
load: 0.54 0.73 0.34
memMB used: 12813
=== 2026-08-26T15:18:20+01:00
57100 47300 47300 47400 54100 57100 49000 (thermal mC)
53.86 W, 53, 95 %
load: 0.85 0.79 0.37
memMB used: 21365
=== 2026-08-26T15:18:50+01:00
73400 58600 58700 59400 73400 71900 61100 (thermal mC)
57.20 W, 63, 96 %
load: 0.96 0.82 0.40
memMB used: 21689
=== 2026-08-26T15:19:20+01:00
79900 64900 65100 65700 79900 78700 67600 (thermal mC)
61.91 W, 70, 96 %
load: 0.98 0.84 0.42
memMB used: 21620
=== 2026-08-26T15:19:50+01:00
83700 67700 67800 68700 83700 79700 70800 (thermal mC)
58.96 W, 70, 96 %
load: 0.99 0.86 0.44
memMB used: 21600
=== 2026-08-26T15:20:20+01:00
85400 70000 70800 70800 85400 82800 73200 (thermal mC)
63.85 W, 73, 96 %
load: 0.99 0.87 0.46
memMB used: 21605
=== 2026-08-26T15:20:50+01:00
87700 71800 72500 72800 87700 83700 75400 (thermal mC)
62.35 W, 76, 95 %

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load: 1.00 0.89 0.48
memMB used: 21552
=== 2026-08-26T15:21:20+01:00
87700 73800 74600 74700 87700 86800 77000 (thermal mC)
64.98 W, 78, 96 %
load: 1.00 0.90 0.50
memMB used: 21556
=== 2026-08-26T15:21:50+01:00
89100 75200 76200 75800 89100 89100 78600 (thermal mC)
64.47 W, 79, 96 %
load: 1.00 0.91 0.51
memMB used: 21577
=== 2026-08-26T15:22:20+01:00
89700 75800 75900 76700 89700 88500 78600 (thermal mC)
66.47 W, 80, 95 %
load: 1.00 0.92 0.53
memMB used: 21548
=== 2026-08-26T15:22:50+01:00
89200 74800 75300 75700 89200 88000 77700 (thermal mC)
64.38 W, 81, 96 %
load: 1.00 0.93 0.55
memMB used: 21540
=== 2026-08-26T15:23:20+01:00
88400 74900 75700 75500 88400 86900 77900 (thermal mC)
65.26 W, 76, 95 %
load: 1.00 0.94 0.56
memMB used: 21571
=== 2026-08-26T15:23:50+01:00
89200 74800 75000 75700 89200 87300 77500 (thermal mC)
65.87 W, 77, 95 %
load: 1.00 0.94 0.58
memMB used: 21557
=== 2026-08-26T15:24:21+01:00
89100 74900 75200 75700 89100 87200 77700 (thermal mC)
62.85 W, 77, 96 %
load: 1.07 0.97 0.60
memMB used: 21556
=== 2026-08-26T15:24:51+01:00
88900 74900 76000 75800 88900 87600 77700 (thermal mC)
66.24 W, 80, 96 %
load: 1.11 0.99 0.62
memMB used: 21575
=== 2026-08-26T15:25:21+01:00
88400 75000 76100 76600 88400 86700 78700 (thermal mC)
66.14 W, 79, 95 %
load: 1.13 1.01 0.64
memMB used: 21530
=== 2026-08-26T15:25:51+01:00
88800 75700 75800 76700 88800 87800 78700 (thermal mC)
62.84 W, 77, 96 %
load: 1.08 1.00 0.65
memMB used: 21544
=== 2026-08-26T15:26:21+01:00
87700 75800 77200 76600 87500 87700 79400 (thermal mC)
63.88 W, 79, 96 %
load: 1.05 1.00 0.66
memMB used: 21543
=== 2026-08-26T15:26:51+01:00
88100 75800 76900 76800 88100 86700 79700 (thermal mC)
63.26 W, 79, 96 %
load: 1.03 1.00 0.67
memMB used: 21555
=== 2026-08-26T15:27:21+01:00
88900 75900 76300 77100 88900 87100 79500 (thermal mC)
64.48 W, 79, 96 %
load: 1.08 1.02 0.69
memMB used: 21600
=== 2026-08-26T15:27:51+01:00
89700 76000 76900 77300 89700 87000 79400 (thermal mC)
65.69 W, 78, 95 %
load: 1.13 1.03 0.71
memMB used: 21557
=== 2026-08-26T15:28:21+01:00
89800 76300 76900 77700 89800 88900 79500 (thermal mC)
63.42 W, 79, 95 %
load: 1.08 1.03 0.72
memMB used: 21610
=== 2026-08-26T15:28:51+01:00
88600 76700 77700 77700 88600 87300 79700 (thermal mC)
63.64 W, 86, 96 %
load: 1.04 1.02 0.73
memMB used: 21579
=== 2026-08-26T15:29:21+01:00
88000 76800 78300 77600 88000 87600 80200 (thermal mC)
65.58 W, 79, 96 %
load: 1.03 1.02 0.74
memMB used: 21576
=== 2026-08-26T15:29:51+01:00
89200 76800 77700 77700 88500 89200 79800 (thermal mC)
66.40 W, 82, 95 %
load: 1.09 1.03 0.75
memMB used: 21552
=== 2026-08-26T15:30:21+01:00

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89000 76800 77800 77800 89000 88400 79800 (thermal mC)
65.53 W, 79, 96 %
load: 1.05 1.03 0.76
memMB used: 21595
=== 2026-08-26T15:30:51+01:00
89500 76900 77700 78300 89200 89500 80100 (thermal mC)
65.48 W, 78, 95 %
load: 1.03 1.03 0.77
memMB used: 21579
=== 2026-08-26T15:31:21+01:00
89500 77000 77400 78200 89500 88600 80600 (thermal mC)
66.60 W, 82, 96 %
load: 1.02 1.02 0.78
memMB used: 21577
=== 2026-08-26T15:31:52+01:00
89500 76900 77400 78400 89500 89300 80700 (thermal mC)
64.59 W, 81, 95 %
load: 1.01 1.02 0.79
memMB used: 21588
=== 2026-08-26T15:32:22+01:00
90000 76900 77700 78600 89800 90000 80100 (thermal mC)
64.09 W, 79, 95 %
load: 1.12 1.05 0.81
memMB used: 21595
=== 2026-08-26T15:32:52+01:00
90700 77100 77500 78500 90100 90700 80500 (thermal mC)
62.66 W, 80, 95 %
load: 1.07 1.04 0.82
memMB used: 21581
=== 2026-08-26T15:33:22+01:00
89400 77000 78200 78500 89400 87900 80700 (thermal mC)
63.63 W, 84, 96 %
load: 1.11 1.05 0.83
memMB used: 21590
=== 2026-08-26T15:33:52+01:00
88500 77400 79000 78500 88400 88500 80700 (thermal mC)
63.52 W, 80, 96 %
load: 1.07 1.05 0.83
memMB used: 21583
=== 2026-08-26T15:34:22+01:00
89600 77500 78400 78600 88500 89600 80000 (thermal mC)
64.72 W, 81, 96 %
load: 1.04 1.04 0.84
memMB used: 21559
=== 2026-08-26T15:34:52+01:00
91000 77500 78400 78600 88900 91000 80700 (thermal mC)
64.96 W, 80, 95 %
load: 1.02 1.04 0.85
memMB used: 21597
=== 2026-08-26T15:35:22+01:00
89100 77400 78500 78600 89000 89100 80700 (thermal mC)
67.86 W, 83, 95 %
load: 1.01 1.03 0.85
memMB used: 21581
=== 2026-08-26T15:35:52+01:00
91300 77700 79100 78600 88200 91300 80700 (thermal mC)
63.66 W, 79, 96 %
load: 1.01 1.03 0.86
memMB used: 21605
=== 2026-08-26T15:36:22+01:00
90600 77600 77700 78700 90000 90600 80400 (thermal mC)
62.50 W, 80, 96 %
load: 1.08 1.04 0.87
memMB used: 21615
=== 2026-08-26T15:36:52+01:00
90200 77700 77800 78700 89800 90200 80200 (thermal mC)
64.03 W, 81, 95 %
load: 1.05 1.03 0.87
memMB used: 21585
=== 2026-08-26T15:37:22+01:00
89500 77800 78000 78700 89500 88000 80700 (thermal mC)
66.61 W, 81, 95 %
load: 1.03 1.03 0.88
memMB used: 21589
=== 2026-08-26T15:37:52+01:00
88900 77700 78800 78600 88900 88600 80700 (thermal mC)
65.27 W, 81, 95 %
load: 1.02 1.03 0.89
memMB used: 21598
=== 2026-08-26T15:38:22+01:00
90300 77700 77800 78700 90000 90300 80500 (thermal mC)
63.74 W, 79, 95 %
load: 1.01 1.02 0.89
memMB used: 21575
=== 2026-08-26T15:38:53+01:00
89000 77800 78600 78600 89000 87900 80400 (thermal mC)
64.41 W, 80, 95 %
load: 1.00 1.02 0.90
memMB used: 21586
=== 2026-08-26T15:39:23+01:00
89800 77800 78900 78700 89800 89000 80600 (thermal mC)
63.79 W, 82, 96 %
load: 1.06 1.03 0.91

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memMB used: 21586
=== 2026-08-26T15:39:53+01:00
89600 78000 79900 78600 88700 89600 80600 (thermal mC)
65.82 W, 82, 95 %
load: 1.04 1.03 0.91
memMB used: 21552
=== 2026-08-26T15:40:23+01:00
89300 77900 79400 78700 89300 88800 80900 (thermal mC)
66.29 W, 81, 95 %
load: 1.02 1.02 0.91
memMB used: 21606
=== 2026-08-26T15:40:53+01:00
91700 77900 81000 78700 89800 91700 81000 (thermal mC)
64.05 W, 79, 96 %
load: 1.09 1.04 0.92
memMB used: 21706
=== 2026-08-26T15:41:23+01:00
91500 77800 78000 79400 91500 91000 81300 (thermal mC)
63.90 W, 83, 95 %
load: 1.45 1.13 0.96
memMB used: 21803
=== 2026-08-26T15:41:53+01:00
91500 77900 77900 79700 91500 90900 81500 (thermal mC)
62.91 W, 83, 96 %
load: 1.67 1.22 0.99
memMB used: 21804
=== 2026-08-26T15:42:23+01:00
91400 78800 80700 79700 91400 89600 81400 (thermal mC)
63.90 W, 80, 96 %
load: 1.85 1.31 1.03
memMB used: 21925
=== 2026-08-26T15:42:53+01:00
89700 77900 77900 79500 89700 89300 81300 (thermal mC)
66.30 W, 79, 95 %
load: 1.57 1.29 1.03
memMB used: 21836
=== 2026-08-26T15:43:23+01:00
90600 77800 78000 79400 89000 90600 81400 (thermal mC)
63.43 W, 82, 96 %
load: 1.34 1.26 1.03
memMB used: 21835
=== 2026-08-26T15:43:53+01:00
90900 77800 78000 78900 88500 90900 81100 (thermal mC)
68.37 W, 83, 95 %
load: 1.28 1.25 1.03
memMB used: 21790
=== 2026-08-26T15:44:23+01:00
90000 77900 79100 79100 89100 90000 81000 (thermal mC)
68.71 W, 83, 95 %
load: 1.17 1.23 1.03
memMB used: 21809
=== 2026-08-26T15:44:53+01:00
91000 77900 78800 79500 90200 91000 81600 (thermal mC)
63.17 W, 82, 95 %
load: 1.10 1.21 1.03
memMB used: 21766
=== 2026-08-26T15:45:23+01:00
92400 78200 80500 79400 88500 92400 80900 (thermal mC)
69.66 W, 83, 95 %
load: 1.06 1.18 1.03
memMB used: 21787
=== 2026-08-26T15:45:54+01:00
92800 78300 80700 79500 88000 92800 81300 (thermal mC)
67.77 W, 86, 96 %
load: 1.03 1.17 1.02
memMB used: 21798
=== 2026-08-26T15:46:24+01:00
90000 78300 79500 79600 89800 90000 81300 (thermal mC)
64.89 W, 80, 95 %
load: 1.44 1.25 1.05
memMB used: 21774
=== 2026-08-26T15:46:54+01:00
92100 78300 80100 79600 89100 92100 81400 (thermal mC)
67.35 W, 82, 96 %
load: 1.66 1.32 1.09
memMB used: 21771
=== 2026-08-26T15:47:24+01:00
90300 78400 81000 79500 88000 90300 81500 (thermal mC)
69.42 W, 83, 95 %
load: 1.72 1.37 1.11
memMB used: 21804
=== 2026-08-26T15:47:54+01:00
90300 78700 79100 79700 90300 89900 81500 (thermal mC)
63.12 W, 80, 96 %
load: 1.70 1.40 1.13
memMB used: 21854
=== 2026-08-26T15:48:24+01:00
91200 79300 82200 80300 89800 91200 81800 (thermal mC)
67.28 W, 82, 96 %
load: 2.07 1.52 1.18
memMB used: 21970

```